# Alternate Interpretations of DDPM

a) DDPM as regression over added "noise".

Recall, the forward process of DDPM,

$$x_t = \sqrt{\bar{\alpha}_t} \cdot x_0 + \sqrt{1-\bar{\alpha}_t} \cdot \epsilon_t \qquad \epsilon_t \sim N(0, I)$$

Rearranging the above terms,

$$x_0 = \frac{x_t - \sqrt{1-\bar{\alpha}_t} \cdot \epsilon_t}{\sqrt{\bar{\alpha}_t}} \quad \text{— (1)}$$

$\epsilon_t$ : noise added to $x_0$, to convert it into $x_t$.

Consistency term in the ELBO : $\| \mu_0 - \mu_q \|_2^2$

$\mu_q$ : mean of $q(x_{t-1} | x_t, x_0)$

$\mu_0$ : mean of $p_0(x_{t-1} | x_t)$

Represent $\mu_q$, interms of $x_t$ & $\epsilon_t$.

$$\mu_q(x_t, x_0) = \frac{\sqrt{\alpha_t} \cdot (1-\bar{\alpha}_{t-1}) \cdot x_t + \sqrt{\bar{\alpha}_{t-1}} \cdot (1-\alpha_t) \, x_0}{1-\bar{\alpha}_t}$$

Replacing $x_0$, above using Eq. 1.

$$\mu_q(x_t, x_0) = \frac{\sqrt{\alpha_t}\cdot(1-\bar{\alpha}_{t-1})\cdot x_t + \sqrt{\bar{\alpha}_{t-1}}\cdot(1-\alpha_t)\left(\frac{x_t - \sqrt{1-\bar{\alpha}_t}\cdot\epsilon_t}{\sqrt{\bar{\alpha}_t}}\right)}{1-\bar{\alpha}_t}$$

($x_0$)

$$\mu_q(x_t, \epsilon_t) = \frac{1}{\sqrt{\alpha_t}}\cdot x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}\cdot\sqrt{\alpha_t}}\cdot\epsilon_t$$

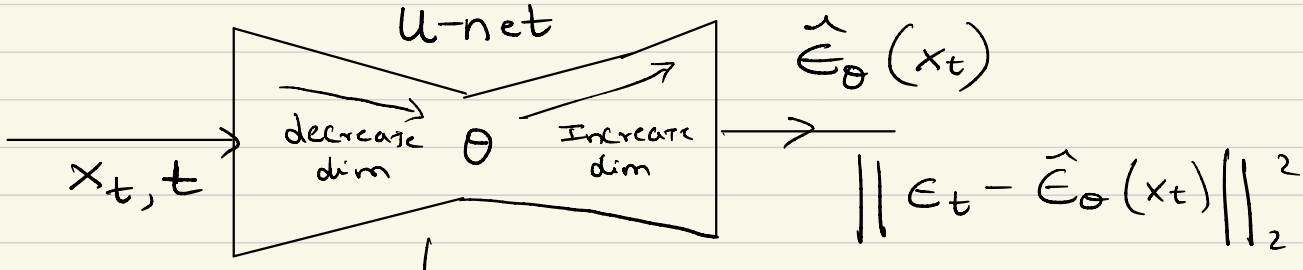
$$\mu_\theta(x_t) = \frac{1}{\sqrt{\alpha}}\cdot x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}\cdot\sqrt{\alpha_t}}\cdot\hat{\epsilon}_\theta(x_t)$$

$$\text{Consistency term} = \frac{1}{2\sigma_q^2}\left\|\mu_q - \mu_\theta\right\|_2^2$$

$$= \frac{1}{2\sigma_q^2}\frac{(1-\alpha_t)^2}{(1-\bar{\alpha}_t)\cdot\alpha_t}\left[\left\|\epsilon_t - \hat{\epsilon}_\theta(x_t)\right\|_2^2\right]$$

$$\propto \left\|\epsilon_t - \hat{\epsilon}_\theta(x_t)\right\|_2^2$$

Regression over the added noise.




↳ Regressor on the added noise

b) DDPM as score-predictors.

suppose $t \sim N(t; \mu_t, \Sigma_t) = p(t)$

Tweedie's formula:

$$\mathbb{E}[\mu_t | t] = t + \Sigma_t \cdot \nabla_t \log p(t)$$

where $\nabla_t \log p(t)$ defined as the "score function"

Recall, that in a DDPM,

$$q(x_t | x_0) = N(x_t; \sqrt{\bar{\alpha}_t} \cdot x_0, (1-\bar{\alpha}_t) I)$$

Apply Tweedie's formula to above,

$$\mathbb{E}(\mu | x_t) = x_t + (1-\bar{\alpha}_t) \cdot \nabla_{x_t} \log p(x_t)$$

The best estimate for $\mathbb{E}(\mu | x_t)$ is the true mean. $\Rightarrow$ $\mathbb{E}(\mu | x_t) = \sqrt{\bar{\alpha}_t} \cdot x_0$.

$$\sqrt{\bar{\alpha}_t} \cdot x_0 = x_t + (1-\bar{\alpha}_t) \cdot \nabla_{x_t} \log p(x_t)$$

Rearranging the terms,

$$x_0 = \frac{x_t + (1-\bar{\alpha}_t) \cdot \nabla_{x_t} \log p(x_t)}{\sqrt{\bar{\alpha}_t}}$$

Expressing the consistency term interms of score-functions.

$$\mu_q(x_t, x_0) = \frac{\sqrt{\alpha_t}\cdot(1-\bar{\alpha}_{t-1})\cdot x_t + \sqrt{\bar{\alpha}_{t-1}}\cdot(1-\alpha_t)\overbrace{\frac{x_t + (1-\bar{\alpha}_t)\cdot\nabla_{x_t}\log p(x_t)}{\sqrt{\bar{\alpha}_t}}}^{x_0}}{1-\bar{\alpha}_t}$$

$$\boxed{\mu_q = \frac{1}{\sqrt{\alpha_t}}\cdot x_t + \frac{1-\alpha_t}{\sqrt{\alpha_t}}\cdot\nabla_{x_t}\log p(x_t)}$$

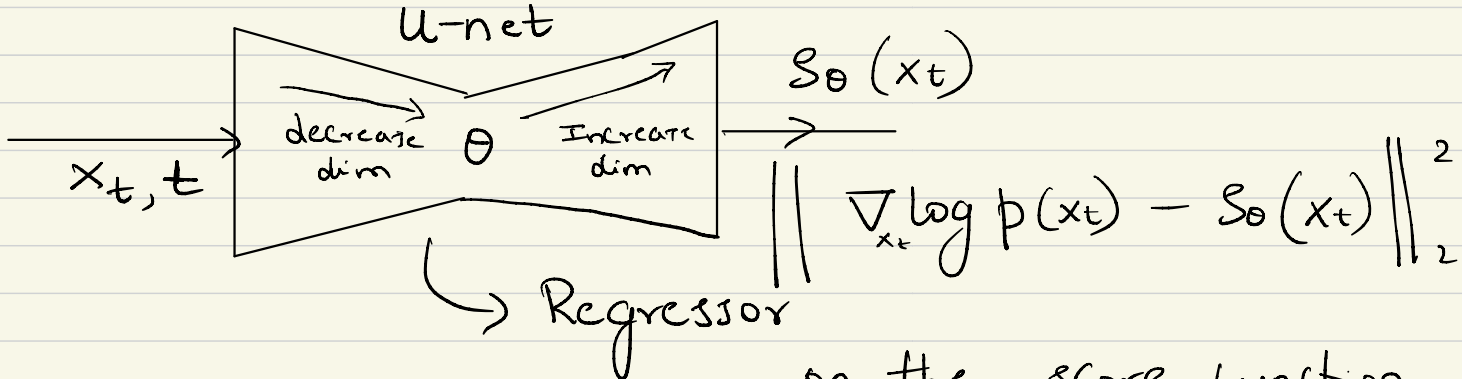
$$\mu_\theta = \frac{1}{\sqrt{\alpha_t}}\cdot x_t + \frac{1-\alpha_t}{\sqrt{\alpha_t}}\cdot s_\theta(x_t)$$

$s_\theta(x_t)$ : predicted score using NN.

consistency term $\propto \|\mu_q - \mu_\theta\|_2^2$

$$\propto \left\| \nabla_{x_t}\log p(x_t) - s_\theta(x_t) \right\|_2^2$$

Regression over the score function.




on the score function.

## To compute the true score

$$x_0 = \frac{x_t + (1-\alpha_t) \cdot \nabla_{x_t} \log p(x_t)}{\sqrt{\alpha_t}} \quad -(1)$$

Also, $x_0 = \frac{x_t - \sqrt{1-\bar{\alpha}_t} \cdot \epsilon_t}{\sqrt{\bar{\alpha}_t}}$ — (2) by definition $\epsilon_t \sim N(0, I)$

Equating (1) & (2), & rearranging

$$\nabla_{x_t} \log p(x_t) = \frac{-1}{\sqrt{1-\bar{\alpha}_t}} \epsilon_t .$$

True score = negatively scaled noise

$\Rightarrow$ a DDPM trained to regress over added noise, is implicitely predicting the score function.

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## Guided Diffusion Models for Conditional Generation.

Data : $(x_0, y)$ where $y$ ; conditioning variable

Eg : $x_0$ : Image $y$ ; Class-label.
$y$ ; text embedding.

Goal : sample from the conditional distribution $p(x_0|y)$ ?

question : How to modify a DDPM to sample from the cond. dist. $p(x_0|y)$ ?

Classifier Guidance :

Recall that DDPMs are score-predictors, predicting the true score of the un-conditional or marginal distribution, $\nabla_{x_t} \log p(x_t)$

From a conditional sampling perspective, the goal is to predict or regress over the conditional score, $\nabla_{x_t} \log p(x_t|y)$

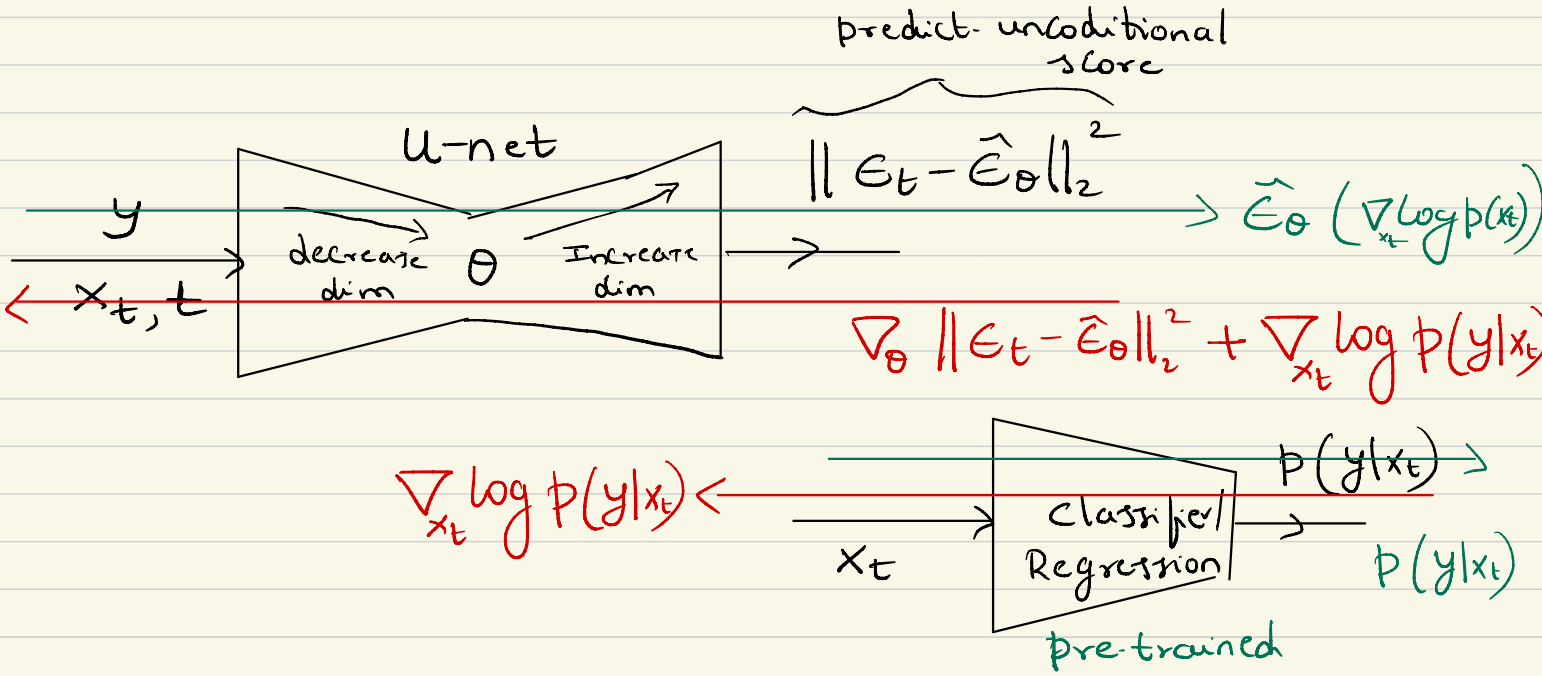
$$\nabla_{x_t} \log p(x_t|y) = \nabla \log \left( \frac{p(x_t) \cdot p(y|x_t)}{p(y)} \right)$$

$$= \nabla_{x_t} \log p(x_t) + \nabla_{x_t} \log p(y|x_t) - \nabla_{x_t} \log p(y)$$

$$\nabla_{x_t} \log p(x_t|y) = \nabla_{x_t} \log p(x_t) + \nabla_{x_t} \log p(y|x_t)$$

unconditional score

Classifier gradient




classifier / Regressor is trained seperately to predict $y$ (class label or Embedding) from $x_t$.

Training the classifier at different noise levels is difficult.

## Classifier - Free Guidance

$$\nabla \log p(x_t|y) = \nabla \log p(x_t) + \nabla \log p(y|x_t)$$

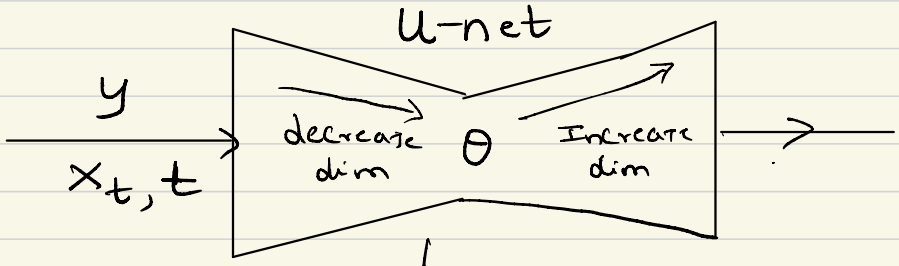
$$\nabla \log p(y|x_t) = \nabla \log p(x_t|y) - \nabla \log p(x_t)$$

↳ Baye's law

$$\nabla \log p(x_t|y) \approx \nabla \log p(x_t) + \alpha \left( \nabla \log p(x_t|y) - \nabla \log p(x_t) \right)$$

$$\boxed{\nabla_{x_t} \log p(x_t|y) = \alpha \nabla_{x_t} \log p(x_t|y) + (1-\alpha) \nabla_{x_t} \log p(x_t)}$$

$\alpha$ : hyper-parameter.

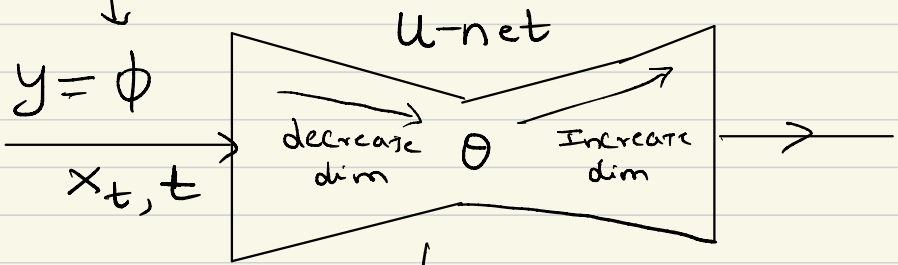



↳ Regressor over $\nabla_{x_t} \log p(x_t|y)$

null label/text

↓

$y = \phi$




(copy of the above)

↳ Regressor over $\nabla_{x_t} \log p(x_t)$

&

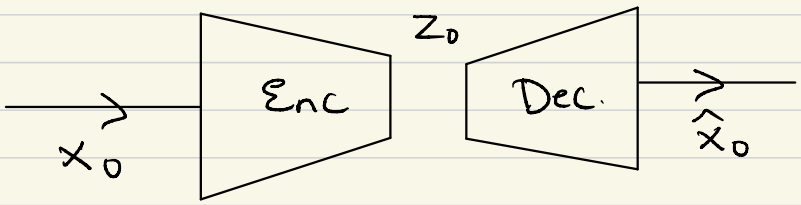
add both the ($\alpha$.)Cond. ^ ($1-\alpha$)UnCond. scores before the gradient step.

# Latent Diffusion Models

The basic idea : Build DMs on the latent space induced by another Encoder-Decoder Models.

Given data $x_0 \in \mathbb{R}^d \sim p(x_0)$, the first step is to learn a latent representation $z_0 \in \mathbb{R}^k$, $k << d$, using an Enc-Dec Model.

Eg : Construct a VQ-VAE to learn a latent space.




Let $E_\phi$, $D_\theta$ represent the Encoder & Decoder functions, respectively.

$E_{\phi^*}$ & $D_{\theta^*}$ are trained using the original data $x_0$ or some other dataset similar to $x_0$

All the datapoints in $\mathbb{R}^d$, are projected on to the latent space as $z_0 = E_{\phi^*}(x_0)$.

A DDPM is constructed on the space of $z_0$.

During inference, to generate a novel point from $p(x_0)$, first sample a novel point from the latent space with reverse diffusion,

$z_{novel}$ & $x_{novel} = D_{\theta^*}(z_{novel}) \sim p(x_0)$.

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## Denoising Diffusion Implicit Models (DDIMs)

DDPMs are slow in sampling ($T = 1000$s) & cannot be used for deterministic posterior Inference.

The forward process is stochastic.

$$x_t = \sqrt{\alpha_t}\, x_0 + \sqrt{1-\alpha_t}\, \epsilon_t \qquad \epsilon_t \sim N(0, I)$$

$\Rightarrow$ Given an $x_0$, running the forward process multiple times leads to different $x_T$ (final latent representation for $x_0$).

$\Rightarrow$ There does not exist a single / unique $x_T$

that generates a given $x_0$, from the reverse diffusion process.

The above issues (slow-sampling & non-unique invertibility) of DDPMs are solved with DDIMs.

DDIMs :

Let us consider a family of inference distributions, parameterized by a vector $\sigma \in \mathbb{R}_{\geq 0}$

$$q_\sigma(x_{1:T}|x_0) \triangleq q_\sigma(x_T|x_0) \prod_{t=2}^{T} q_\sigma(x_{t-1}|x_t, x_0)$$

where $q_\sigma(x_T|x_0) \triangleq \mathcal{N}(\sqrt{\alpha_T}\, x_0, (1-\alpha_T)\, I)$ &

for all $t > 1$

$$q_\sigma(x_{t-1}|x_t, x_0) \triangleq \mathcal{N}\left(\sqrt{\alpha_{t-1}}\, x_0 + \sqrt{1-\alpha_{t-1}-\sigma_t^2} \cdot \frac{x_t - \sqrt{\alpha_t}\, x_0}{\sqrt{1-\alpha_t}},\ \sigma_t^2 I\right)$$

The forward process corrs. to the above posterior is :

$$q_\sigma(x_t|x_{t-1}, x_0) = \frac{q_\sigma(x_{t-1}|x_t, x_0)\, q_\sigma(x_t|x_0)}{q_\sigma(x_{t-1}|x_0)} \quad \text{Bayes' rule.}$$

The above forward process is not Markovian.

$\Rightarrow \quad x_t \not\perp x_0 \mid x_{t-1}$.

In the above process,

$$q_o(x_t|x_0) = \mathcal{N}\left(x_t ; \sqrt{\alpha_t}\cdot x_0, \sqrt{1-\alpha_t}\cdot I\right)$$

the same as DDPM.

Since, $q_o(x_t|x_0)$, conditional of $x_t$ given $x_0$ is the same in both DDPM & DDIM, the ELBO for both of them are the same & optimizing the ELBO leads to the same solution.

∴ the ELBO in DDPM | DDIM depends only on $q(x_t|x_0)$

$\Rightarrow$ as long as $q(x_t|x_0)$ is same as that of DDPMs, any process can be used & leads to the same solution for ELBO.

as far as $q(x_t|x_0)$ is same as DDPM, any definition for $q(x_{1:T}|x_0)$ can be chosen.

## Defining the Decoder or the Reverse process

$$p_\theta(x_{t-1}|x_t) = \begin{cases} \mathcal{N}\left(\mu_\theta(x_1), \sigma^2 I\right) & t=1 \\ q_\sigma\left(x_{t-1}|x_t, \mu_\theta(x_t)\right) & \text{otherwise.} \end{cases}$$

where $\mu_\theta(x_t) = \dfrac{x_t - \sqrt{1-\alpha_t}\cdot\hat{\epsilon}_\theta}{\sqrt{\alpha_t}}$ : approx. for $x_0$

Recall, $x_t = \sqrt{\alpha_t}\cdot x_0 + \sqrt{1-\alpha_t}\cdot\epsilon_t$ , $\epsilon_t \sim \mathcal{N}(0, I)$

$$x_0 = \frac{\left(x_t - \sqrt{1-\alpha_t}\cdot\epsilon_t\right)}{\sqrt{\alpha_t}}$$

The main result: The ELBO corresponding to the above non-markovian family of models & that of DDPM are the same (off-by a const.)

$\Rightarrow$ Training the regression network in DDPM is implicitly training all the above non-Markovian Models.

There is no difference b/w training DDPM & DDIMs, but the inference process is different.

# Inference with DDIMs.

Sampling from $p_\theta(x_{t-1}|x_t)$

$$p_{\theta^*}(x_{t-1}|x_t) = \begin{cases} \mathcal{N}(\mu_{\theta^*}(x_1), \sigma^2 I) & t=1 \\ q_\sigma(x_{t-1}|x_t, \mu_{\theta^*}(x_t)) & \text{otherwise.} \end{cases}$$

$$\mu_\theta(x_t) = \frac{x_t - \sqrt{1-\alpha_t}\cdot\hat{\epsilon}_\theta}{\sqrt{\alpha_t}} \Rightarrow \frac{x_t - \sqrt{\alpha_t}\cdot\mu_{\theta^*}}{\sqrt{1-\alpha_t}} = \hat{\epsilon}_\theta$$

$$q_\sigma(x_{t-1}|x_t, \mu_{\theta^*}) \triangleq \mathcal{N}\left(\sqrt{\alpha_{t-1}}\,\mu_{\theta^*} + \sqrt{1-\alpha_{t-1}-\sigma_t^2}\cdot\frac{x_t - \sqrt{\alpha_t}\,\mu_{\theta^*}}{\sqrt{1-\alpha_t}},\ \sigma_t^2 I\right)$$

putting all the above together, sampling from $p_{\theta^*}(x_{t-1}|x_t)$ is as follows:

$$z \sim \mathcal{N}(0, I)$$

$$x_{t-1} = \sqrt{\alpha_{t-1}}\left[\frac{x_t - \sqrt{1-\alpha_t}\cdot\hat{\epsilon}_{\theta^*}}{\sqrt{\alpha_t}}\right] + \sqrt{1-\alpha_{t-1}-\sigma_t^2}\cdot\hat{\epsilon}_{\theta^*} + \sigma_t^2\cdot z$$

when $\sigma_t = \frac{\sqrt{(1-\alpha_{t-1})}}{\sqrt{1-\alpha_t}} \cdot \frac{\sqrt{1-\alpha_t}}{\sqrt{\alpha_{t-1}}}$,

the above random process reduces to DDPM.

when $\sigma_t = 0$, the sampling process has the following form:

$$x_{t-1} = \sqrt{\alpha_{t-1}} \left[ \frac{x_t - \sqrt{1-\alpha_t} \cdot \hat{\epsilon}_\theta}{\sqrt{\alpha_t}} \right] + \sqrt{1-\alpha_{t-1}} \cdot \hat{\epsilon}_\theta$$

The above sampling process is deterministic.

The only randomness is from $x_T \sim N(0, I)$

with $\sigma_t = 0$, the above model is called DDIM.

$\Rightarrow$ the forward process in DDIM with $\sigma_t = 0$, is also deterministic.

The above implies that DDIM has a unique invertibility. (DDIM Inversion).

End of Diffusion Models